Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2015 JC2 Preliminary Examinations

H2 BIOLOGY **9648/1**

Paper 1: Multiple Choice

Friday 18 September 2015 1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **40** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

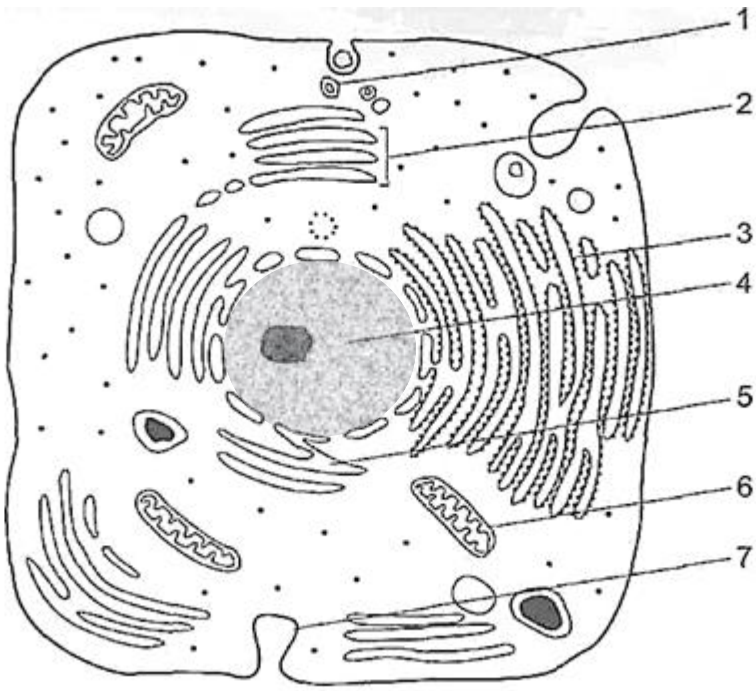
Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet only.

1 ~~Figure below~~The figure below shows ~~is~~ an electron micrograph of a human cell.



Which of the following does not correctly identify its structure-function adaptation?

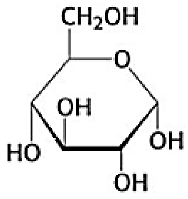
	Function	Structural adaptation
A	Cholesterol production	Extensive **5**
B	Enzyme secretion	Extensive **1**, **2, 3** and **6**
C	Phagocytosis	Extensive **1** and **6**
D	Absorption	Extensively-folded **7**

2 Which of the following processes do not involve lysosomes?

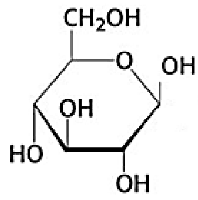
1 Cell division
2 Intracellular digestion
3 Protein synthesis
4 Removing damaged mitochondria

A 1 and 3 only
B 2 and 4 only
C 3 and 4 only
D 1, 2 and 4 only

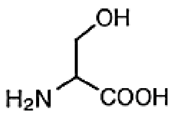
3 Figure below shows the structure of four monomers.



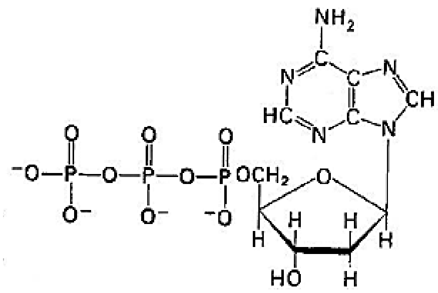
U



X



Y



Z

Which of the following combination of polymer, monomer and bond formed between monomers is correct?

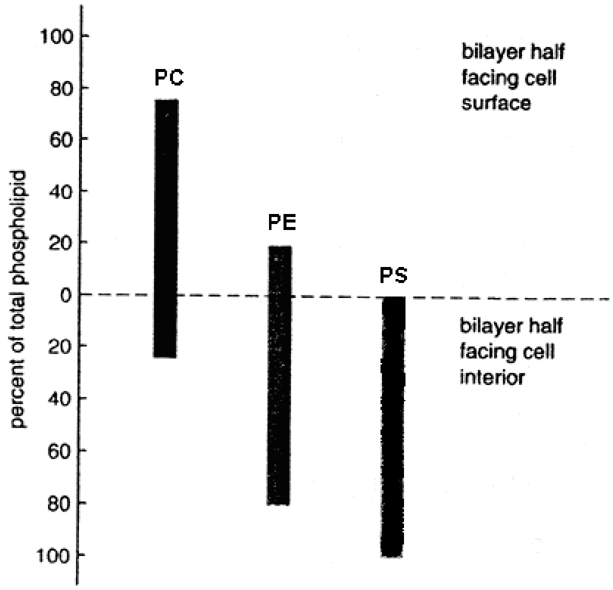
	starch	*cellulose*	*polypeptide*	*polynucleotide*
A	**X**, β-1,4 glycosidic bond	**U**, α-1,4 glycosidic bond	**Z**, ester linkage	**Y**, disulphide linkage
B	**U**, α -1,4 glycosidic bond	**X**, β -1,4 glycosidic bond	**Y**, peptide bond	**Z**, phosphoester linkage
C	**Z**, peptide bond	**X**, hydrogen bond	**Z**, ionic bond	**U**, hydrogen bond
D	**X**, ionic bonds	**Y**, peptide bond	**U**, hydrogen bond	**Z**, α -1,6 glycosidic bond

4 Which of the following statement(s) describe(s) unsaturated fats?

1 They have a higher melting point than saturated fats.
2 Each molecule is an ester formed by a fatty acid with an alcohol other than glycerol.
3 At room temperature, they are closely packed.
4 They contain double bonds that cause kinks in the hydrocarbon chain.

A 4 only
B 2 and 3 only
C 1 and 4 only
D 2, 3 and 4 only

5 The following diagram illustrates the types of membrane phospholipids found in the intact erythrocytes (red blood cells).



The following membrane phospholipids are found in the erythrocyte membranes:

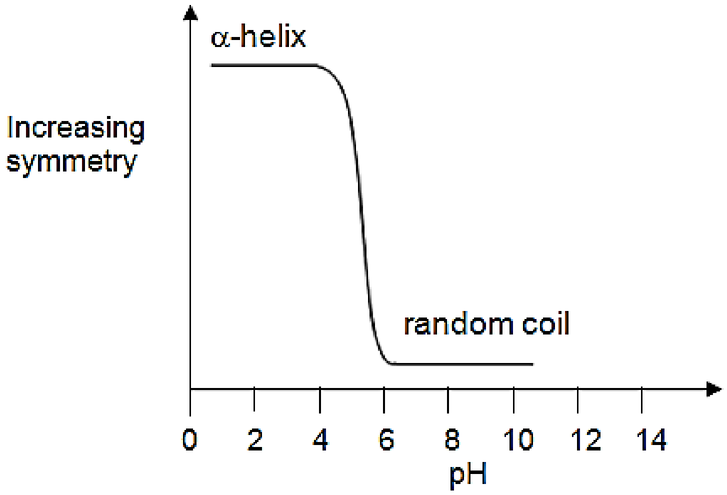
PC = phosphatidyl choline
PE = phosphatidyl ethanolamine
PS = phosphatidyl serine

Which of the following statement(s) correctly identifies with the graph?

1 76% of the total membrane phospholipids contain choline and 20% contain ethanolamine.
2 24% of the inner total membrane phospholipids contain choline and 80% ethanolamine.
3 Most **PC** is confined to the outer surface of the erythrocytes while most of the **PE** are confined to the inner surface of the erythrocytes.
4 Membranes, in general, can be concluded to be asymmetric.

A 1 and 2 only
B 2 and 3 only
C 1 and 4 only
D 2 and 4 only

6 The graph below shows ~~Figure below is a graph showing~~ the effect of pH on the structure of a protein which consists of entirely of repeating residues of one amino acid.



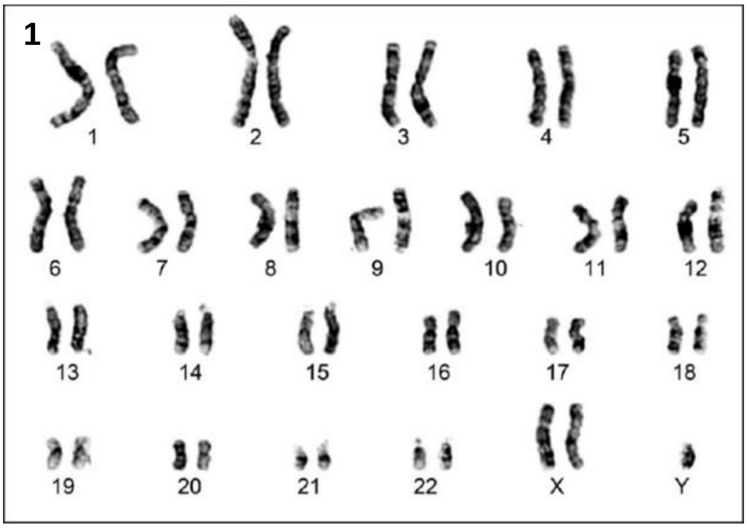
Which statement is true?

A At low acidity the protein loses its primary structure.
B At low acidity the protein loses its secondary structure.
C At high acidity the protein loses its secondary structure.
D At high acidity the protein loses its tertiary structure.

7 The Thompson seedless grape is triploid, with three copies of each chromosome. Which phase of the cell cycle, when disrupted, would result in such a mutation?

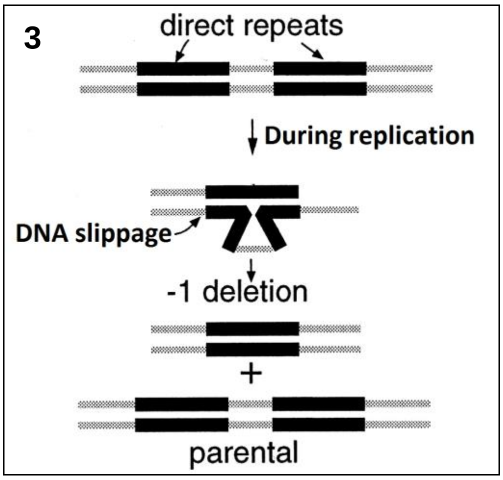
A Anaphase
B Anaphase I
C Anaphase II
D Cytokinesis

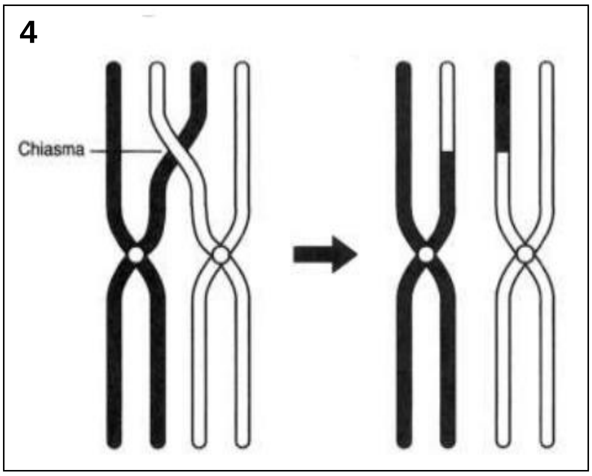
8 Which of the following diagrams 1 to 4 ~~is~~ does **not** ~~an example of~~show chromosomal mutation?



2

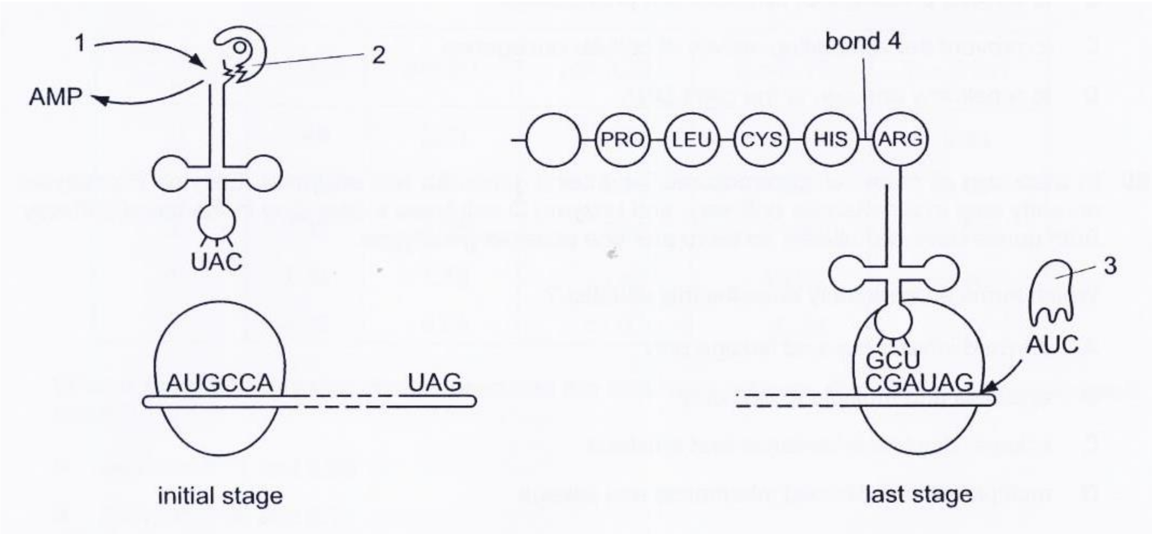
Normal allele	...A C T C C T G A G G A G...
Mutated allele	...A C T C C T G T G G A G...





A 1 and 2
B 1 and 3
C 2 and 4
D 2, 3 and 4

9 A number of molecules other than tRNA and mRNA are involved during translation. The diagram below shows some of these molecules and some of the nucleotides in the codon and anticodon positions.



Which line in the table is correct for labels 1 – 4?

	1	2	3	4
A	ADP	Aminoacyl tRNA synthetase	Amino acid	Hydrogen bond
B	ADP	Amino acid	Translation releasing factor	Hydrogen bond
C	ATP	Amino acid	Aminoacyl tRNA	Peptide bond
D	ATP	Aminoacyl tRNA synthetase	Releasing factor	Peptide bond

10 Which correctly describes the structure of DNA during transcription?

A Double-stranded region with two loops of unpaired bases.
B Double strand held together by covalent bonds between nucleotide pairs.
C Double-stranded with single-stranded sections associated with polymerase enzymes.
D Single strand formed by base pairing cytosine with guanine and adenine with thymine.

11 After sequencing of the fragment carrying a mutation in the gene coding for phosphofructokinase, four changes were observed:

mutation	*wild-type sequence*	*mutant sequence*
W	AUU	GUU
X	GAU	GAA
Y	GAC	GAU
Z	UUU	UCU

With reference to Tables 11.1 and 11.2, which of the mutations (**W**, **X**, **U** or **Z**) would you test immediately to confirm that this is the relevant mutation that resulted in a change in the tertiary structure of the phosphofructokinase?

Second letter

First letter	U	C	A	G	Third letter
U	UUU UUC } Phe UUA UUG } Leu	UCU UCC UCA UCG } Ser	UAU UAC } Tyr UAA Stop UAG Stop	UGU UGC } Cys UGA Stop UGG Trp	U C A G
C	CUU CUC CUA CUG } Leu	CCU CCC CCA CCG } Pro	CAU CAC } His CAA CAG } Gln	CGU CGC CGA CGG } Arg	U C A G
A	AUU AUC AUA } Ile AUG Met	ACU ACC ACA ACG } Thr	AAU AAC } Asn AAA AAG } Lys	AGU AGC } Ser AGA AGG } Arg	U C A G
G	GUU GUC GUA GUG } Val	GCU GCC GCA GCG } Ala	GAU GAC } Asp GAA GAG } Glu	GGU GGC GGA GGG } Gly	U C A G

Table 11.1

Classification of the amino acids based on side chain reactivity and polarity at pH 7.4

Group I, Hydrophobic	Group II, Polar, Uncharged	Group III, Polar, Charged
Gly	Ser	Asp
Ala	Thr	Glu
Val	Cys	Lys
Leu	Tyr	Arg
Ile	Asn	
Pro	Gln	
Met	His	
Phe		

Table 11.2

A **W**
B **X**
C **Y**
D **Z**

12 Which statements about typical plasmids in bacterial conjugation are correct?

1 An F plasmid is a circular, double-stranded molecule of DNA.
2 One strand of the F plasmid is nicked at the origin of replication.
3 An F plasmid DNA strand enters the recipient cell beginning at its 3’ end.
4 After transfer of F plasmid DNA, complementary strands of F plasmid DNA are synthesized in both donor and recipient cells.
5 F plasmid DNA is made into a circle by the action of DNA ligase.

A 1, 2 and 3
B 1, 2 and 4
C 1, 4 and 5
D 3, 4 and 5

13 Some features of the life cycle of a virus are listed below.

1 Fusion between viral envelope and host cell plasma membrane
2 Integration with host DNA
3 Replication of viral nucleic acid
4 Formation of cDNA
5 Assembly of new virus particles

Which virus has all of these features in its reproductive cycle?

A HIV
B Influenza virus
C lambda phage
D T4 phage

14 Which feature is common in eukaryotic and prokaryotic genomes?

A Both genomes contain non-coding sequences.
B Both require general transcription factors to initiate transcription.
C Both genomes are replicated during interphase just before binary fission or mitosis.
D The DNA in both genomes can be methylated to prevent hydrolysis by their own restriction enzymes.

15 An analysis of the cells of a cancer patient revealed the presence of abnormal amounts of *Ras* proteins. Which of the following could not explain for this observation?

A Gene amplification of *Ras* proto-oncogene.
B A point mutation in a control element of the *Ras* proto-oncogene.
C Translocation of *Ras* proto-oncogene to a region under the control of a more active promoter.
D A single substitution in the *Ras* proto-oncogene which results in a constitutively active *Ras* protein.

16 The active messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenised cell contents through a fractionating column. The column has short lengths of uracil nucleotides attached to a solid supporting material. Most molecules of mRNA that pass through the column break up into small pieces and cannot be translated.

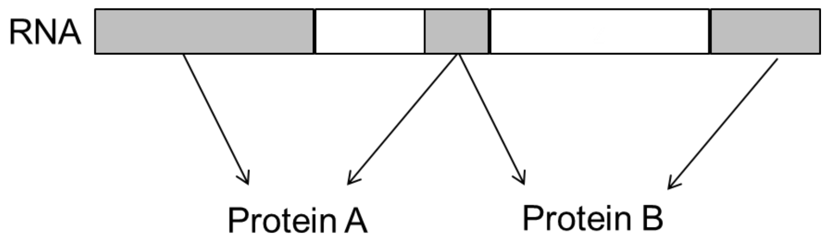
The active mRNAs that attach to the column can be separated by appropriate treatment.

Which statements correctly describe active mRNA?

1 Active mRNAs are held to the fractionating column by bonds between adenine and uracil bases.
2 Active mRNAs can be released from the fractionating column by breaking hydrogen bonds.
3 Only mRNAs with polyadenine tailing can be translated.

A 1 only
B 1 and 2
C 2 and 3
D 1, 2 and 3

17 The diagram below shows that RNA transcribed from a length of DNA of a chromosome codes for two different proteins.



Which of the following statement is correct?

A The DNA template transcribed to form this RNA was part of a eukaryotic chromosome because introns have been edited out of the RNA.
B The DNA template transcribed to form this RNA was part of a eukaryotic chromosome because this is a way of saving space in a small genome.
C The DNA template transcribed to form this RNA was part of a prokaryotic chromosome because introns have been edited out of the RNA.
D The DNA template transcribed to form this RNA was part of a prokaryotic chromosome because this is a way of saving space in a small genome.

18 Which of the following statement about cancer is true?

A Tumours invade other tissues due to the loss of density-dependent inhibition.
B Cancer is a result of decreased cell death due to the mutation of a proto-oncogene.
C Anchorage dependence and the ability to undergo differentiation are characteristics of a cancerous cell.
D Individuals who inherit one mutant copy of tumour suppressor gene are as likely to develop cancer as individuals with two normal copies.

19 The length of the petiole (leaf stalk) in a type of flowering plant is controlled by two genes, A and B. These genes are found on different loci on non-homologous chromosomes.

Homozygous dominant plants have long petioles (30 cm), homozygous recessive plants have short petioles (10 cm). Each dominant allele contributes 5cm to the petiole length.

F_1 plants with medium length petioles (20 cm) were obtained when a plant with short petiole is crossed with a plant with long petiole. If the F1 generation plants were allowed to cross, what proportion of their offspring would be expected to have medium length (20 cm) petioles?

A 0.0625
B 0.25
C 0.375
D 0.5

20 In a species of mammal, chromosome 7 contains genes for two enzymes. Enzyme **P** catalyses an early step in a metabolic pathway, and enzyme **Q** catalyses a later step in the same pathway. Both genes have two alleles, so there are nine possible genotypes.

Which terms appropriately describe this situation?

A dihybrid inheritance and linkage only
B epistasis and multiple alleles only
C linkage, dihybrid inheritance and epistasis
D multiple alleles, dihybrid inheritance and linkage

21 The Himalayan rabbits have white hair on the body and black hair on the extremities such as feet, tail, ears and face. The allele for the Himalayan rabbit pigment pattern, c^h, is recessive to the alleles for normal colour (all hair agouti), C, as well as dark chinchilla (all hair dark grey), c^{chd}, and is dominant to the allele for albino (all hair white, no pigment production), c. All of the alleles of this gene produce different versions of the same enzyme involved in pigment production.

A patch of white fur was removed from a Himalayan rabbit and an ice pack secured to the skin. The fur that grew back on the patch was black.

Which is correct?

	Genotypes of Himalayan rabbits	**Explanation for pigment pattern in Himalayan rabbits**
A	c^hc^h only	The enzyme is denatured at the high skin temperatures found on the rabbit's bodies
B	c^hc^h only	The enzyme becomes inactive at the low skin temperatures found on the rabbit's feet, tail, ears and face.
C	c^hc^h and c^hc only	The enzyme is denatured at the high skin temperatures found on the rabbit's bodies
D	c^hc^h and c^hc only	The enzyme becomes inactive at the low skin temperatures found on the rabbit's feet, tail, ears and face.

22 Lactate dehydrogenase (LDH) is an important enzyme in mammalian skeletal muscles where oxygen can be in short supply. The enzyme catalyzes the following reversible reaction:

$$\text{pyruvate + NADH} \underset{\text{LDH}}{\rightleftharpoons} \text{lactate + NAD}^+$$

Which of the following is false?

A NADH can be re-oxidised.
B ATP production can continue.
C More glucose can be broken down.
D Lactate acts as a hydrogen acceptor.

23 Which of the following statement(s) is/are true regarding cyclic and non-cyclic photophosphorylation?

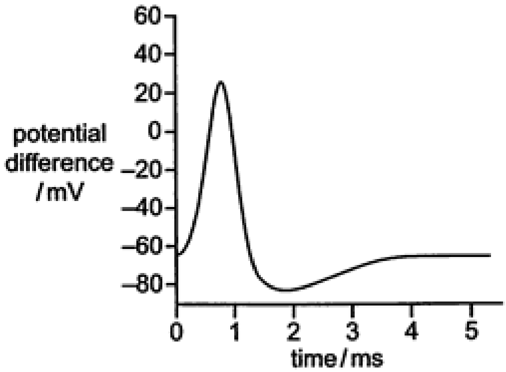
1 Only cyclic photophosphorylation produces oxygen.
2 Only cyclic photophosphorylation can function in the absence of photosystem II.
3 Only non-cyclic photophosphorylation will be affected in the absence of NADP reductase.
4 The plant switches from cyclic to non-cyclic photophosphorylation when only ATP is required.

A 3 only
B 1 and 4 only
C 2 and 3 only
D 2 and 4 only

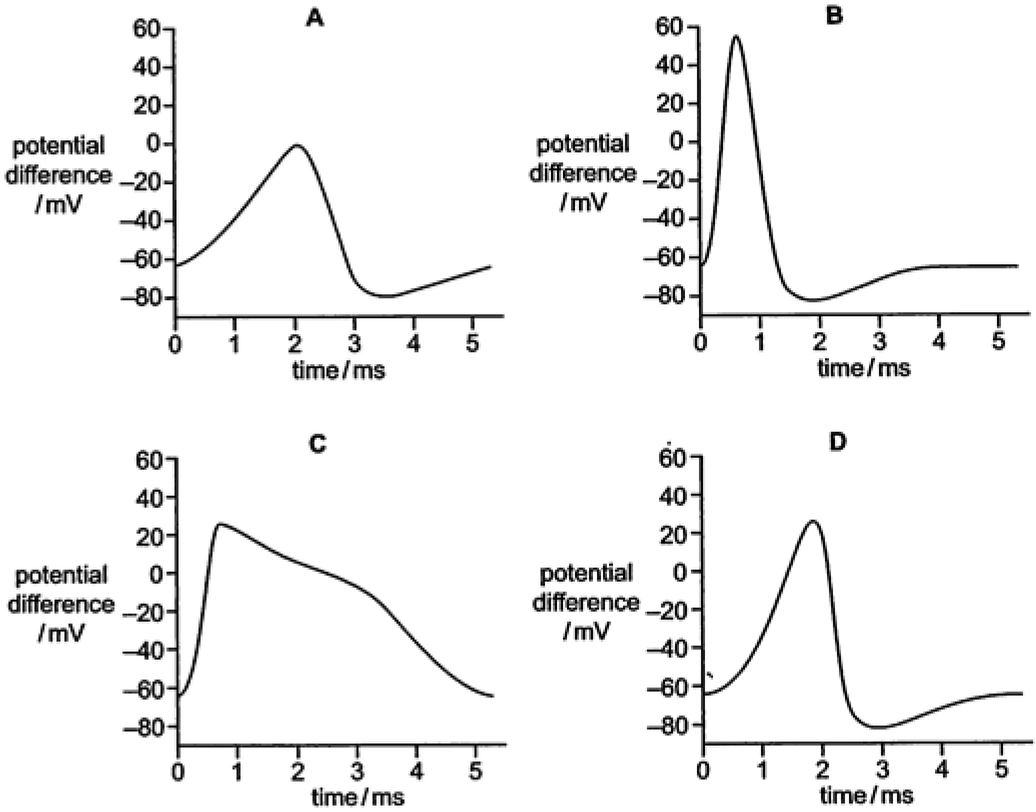
24 Which feature of the transmission of nerve impulses does **not** have a role in maintaining the unidirectional movement of that signal?

A Ligand-gated Na^+ channels on the postsynaptic membrane.
B Exocytosis of synaptic vesicles at the presynaptic membrane.
C Neurotransmitter-hydrolysing enzyme located on the postsynaptic membrane.
D Neurotransmitter diffuses down concentration gradient from presynaptic neurone to postsynaptic neurone.

25 The diagram below shows a normal action potential.



Some drugs can bind and block voltage-gated K^+ channels. Which of the following diagrams shows an action potential in a neurone affected by this group of drugs?



26 What is necessary for an organism to respond to changes in its internal environment?

A A communication system between receptors and effectors.
B A communication system involving nerve cells.
C Negative feedback from receptor to effector.
D Positive feedback from effector to receptor.

27 In a clinical trial, the effectiveness of two types of synthetic drug for a certain disease was tested. Participants in this test were divided into two groups. One group received Drug A. The second group received Drug B. All participants received the same amount and concentration of the appropriate drug.

The table below shows the average blood glucose levels after injection of the drug for both groups of participants after a standardised meal. Normal blood glucose level for a healthy person is 90 mg / 100 ml.

Time after injection (h)	Average blood glucose level in participants injected with Drug A (mg / 100 ml)	Average blood glucose level in participants injected with Drug B (mg / 100 ml)
0	360	365
1	231	326
2	180	271
3	160	222
4	160	152
5	160	120
6	160	90

Which of the following statements is **not** true?

1 Drug A is useful for Type II diabetes.
2 Drug A may be degraded roughly 3 hours after injection.
3 Drug A must be injected more frequently than drug B in a day.
4 Drug B resembles a hormone released from the beta cells of the islets of Langerhans in the pancreas.

A 1 only
B 4 only
C 1 and 2
D 2 and 3

28 A scientist is investigating the effects of Poison T on the cell signaling pathway of glucagon. It is found that Poison T diminishes the effect of glucagon.

The table below shows the different components of the cell signaling pathway of glucagon and their statuses.

Component	**Status**
G-protein coupled receptor (GPCR)	Configuration can be changed
G-protein	Can exchange GDP for GTP
cAMP levels	low
Protein kinase A	Inactivated

Which of the following statements can be concluded from the results?

1 Poison T interferes with reception.
2 Poison T prevents G-protein from hydrolysing GTP.
3 Poison T inactivates the enzyme adenylate cyclase.
4 Poison T stops the relaying of message down the phosphorylation cascade.

A 1 and 2
B 2 and 3
C 2 and 4
D 3 and 4

29 The huia, *Heteralocha acutirostris*, was found in New Zealand until 1907, when it became extinct. This bird had a ground-feeding habit and was particularly noted for large, attractive tail feathers. Males and females had very different beak forms, with the males having a short strong beak, whilst the females had a long curved beak to reach into otherwise inaccessible places.

What is the most likely reason for the extinction of the huia?

A Humans introduced many species which preyed upon the huia.
B New competitors in New Zealand occupied part of the huia’s niche.
C Male and female huia were unable to breed successfully owing to strong sexual dimorphism.
D A few huia became geographically isolated when they arrived at another island, leading to their reduced genetic variation.

30 Which of the following descriptions refers accurately to homologous structures?

	Basic structure	**Common ancestor**	**Function performed**
A	Same	No	Different
B	Different	No	Same
C	Same	Yes	Different
D	Different	Yes	Same

31 Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome C. Cytochrome C is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the ideas that all living organisms are related and that there is a single, rather than a multiple, origin of life?

1 The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
2 The sequence of amino acids in cytochrome C is similar in organisms that are from similar environments or with similar metabolic demands.
3 The majority of organisms have the same, or similar, amino acid sequences for cytochrome C.
4 When transferred into a very dissimilar organism, a gene coding for cytochrome C will lead to the expression of a protein that will function in the other organism.

A 1 and 2
B 2 and 3
C 3 and 4
D 1, 3 and 4

32 If the first three nucleotides in a six-nucleotide restriction site are ATC, what would the next three nucleotides most likely be?

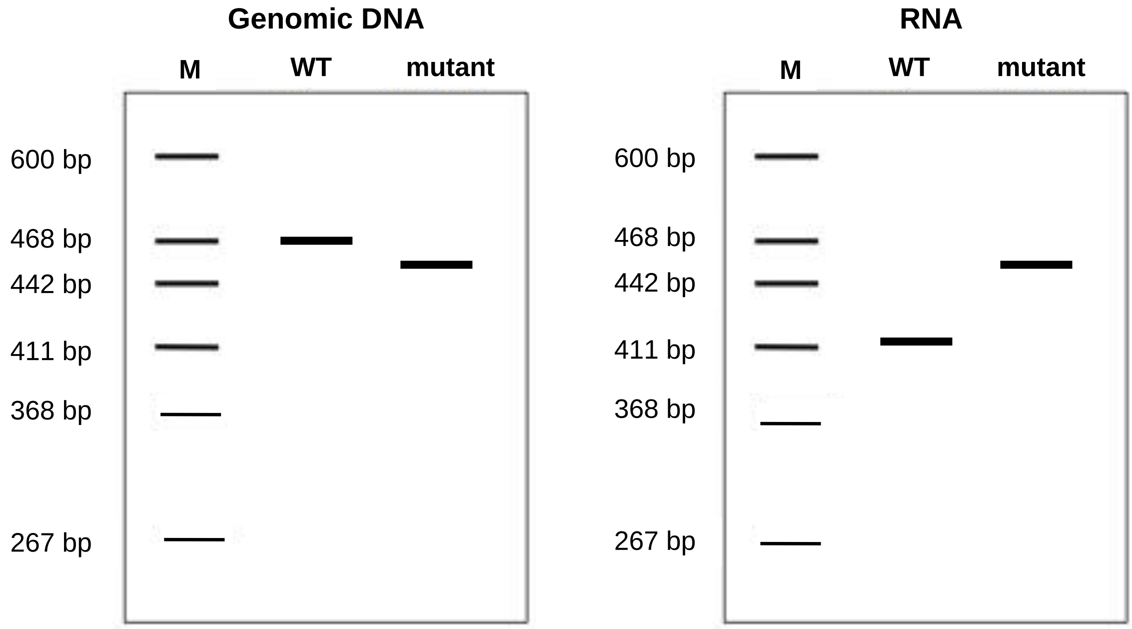
A ATC
B CTA
C GTG
D GAT

33 Which features apply to cDNA libraries?

1 Represent mRNA from a particular cell or tissue.
2 Useful in the study of genetic mutations in cancer cells.
3 Useful for cloning of DNA for in vitro studies of gene function.
4 Gene probing and X-ray autoradiography are necessary for producing cDNA libraries.

A 1 and 3
B 2 and 4
C 1, 2 and 3
D 1, 3 and 4

34 In an investigation of a gene suspected to be involved in a genetic disease, separate PCR procedures were done using genomic DNA and RNA isolated from healthy (wild-type WT) and diseased cells (mutant). The PCR products were analysed on polyacrylamide gels. The sizes of the bands in the molecular weight marker (M) are indicated in base-pairs.



Which of the following best explains the results obtained?

A Deletion of an exon in the mutant.
B Deletion of a splice site in the mutant.
C Deletion of several codons in the mutant.
D Deletion of several introns in the mutant.

35 Which is a correct statement about obtaining human embryonic stem cells for research?

1 Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.
2 The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.
3 The cells must be removed within a day following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.
4 The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.

A 2 only
B 1 and 2
C 2 and 3
D 3 and 4

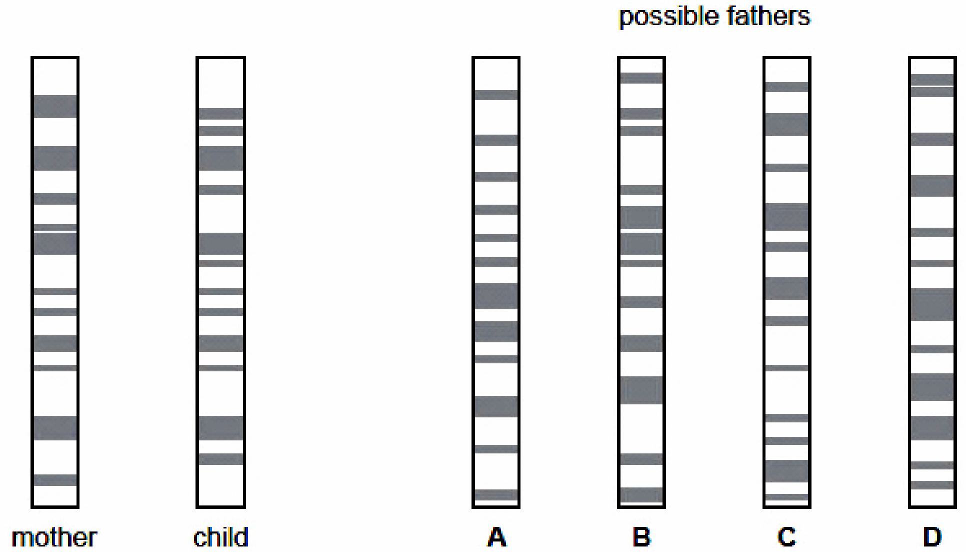
36 It is possible to introduce an allele for a functioning CFTR protein into lung epithelial cells of patients suffering from the genetically inherited condition cystic fibrosis.

Why can this strategy never provide a permanent cure for the patient?

A this is only somatic and not germ line therapy
B epithelial cells are continually dying and being replaced
C the methods of inserting the allele have low success rate
D the DNA molecule that makes up the functioning allele is very unstable

37 DNA fingerprinting can be used to determine the paternity of a child. DNA from the mother and the child is cut into fragments, separated by electrophoresis and made visual using a stain. The diagram shows the genetic profiles of a mother and child, and four possible fathers.

Who is the father?



38 The human genome project has identified and mapped the genes on human chromosomes. This is allowing scientists to identify specific, faulty genes which contribute to inherited conditions.

This is useful in many ways, for example

1 carriers of faulty genes can be advised about changes in lifestyle to minimise risks.
2 carriers of faulty genes can be identified and informed of their risk status.
3 diagnostic tests can be developed to identify carriers of faulty genes.
4 drugs can be developed to block the action of problem genes.
5 embryos can be screened to avoid the birth of affected children.
6 employers can take account of the genetic predisposition of employees.

Which two uses arise **directly** from the information provided by the project?

A 1 and 2
B 2 and 5
C 3 and 4
D 5 and 6

39 What explains why explants from leaves are grown on a nutrient medium containing carbohydrate, mineral nutrients and vitamins?

A The callus is undifferentiated and cannot photosynthesise.
B The carbon source is needed to stimulate callus formation.
C The medium switches off genes that stimulate cell division.
D Vitamins allow the medium to be osmotically balanced.

40 Which genetic modification would enable higher yield in crops grown on fertile soil in a tropical region?

A salt tolerance
B drought tolerance
C increased carbon fixation
D increased vitamin A content